IMPORTANT **Definition of automatic relevance determination (ARD)**

Using a kernel with a length scale for each dimension is called automatic relevance determination, or ARD. This term denotes the fact that after using gradient descent to optimize these length scales, we can infer how relevant each dimension of the objective function is with respect to the function values. A dimension with a large length scale has low variability and is therefore less relevant in modeling the objective function values than a dimension with a small length scale.

Implementing ARD is very easy to do with GPyTorch: we simply specify the `ard_num_dims` parameter to be equal to the number of dimensions that our objective function has when initializing our covariance function. This is done with the RBF kernel like so:

```
class ARDGPModel(gpytorch.models.ExactGP):
    def __init__(self, train_x, train_y, likelihood):
        super().__init__(train_x, train_y, likelihood)
        self.mean_module = gpytorch.means.ZeroMean()
        self.covar_module = gpytorch.kernels.ScaleKernel(
            gpytorch.kernels.RBFKernel(ard_num_dims=2)
        )

    def forward(self, x):
        ...  ❶
```

❶ Omitted.

Let's see whether, when trained on our Ackley function, this model gives us different length scales for the two dimensions. To do this, we first construct a randomly sampled training data set consisting of 100 points:

```
torch.manual_seed(0)
train_x = torch.rand(size=(100, 2)) * 6 - 3
train_y = ackley(train_x)
```

After training the model using gradient descent like we have been doing, we could inspect the optimized values for the length scales by printing out:

```
>>> model.covar_module.base_kernel.lengthscale
tensor([[0.7176, 0.4117]])
```

This is indeed the result that we expected: a larger length scale for the first dimension, where there is less variability in the function values, and the smaller length scale for the second dimension, where there is more variability.

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